



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of
Computing

Proposal:

Student Event Registration System

Programme	: Bachelor of Computer Science (<i>Data Engineering</i>)
Subject Code	: SECJ2154
Subject Name	: Object Oriented Programming (<i>OOP</i>)
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Prepared by : LAU YEE WEN (A23CS0099)

NURUL ASYIKIN BINTI (A23CS0162)

KHAIRUL ANUAR

EHTASHAM AL NOMAN (A22EC4007)

Section : 02

Lecturer : DR ZURAINI BINTI ALI SHAH

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1.0 Introduction

Student organizations and university faculties often conduct a variety of events, including workshops, lectures, competitions, and cultural performances. Traditionally, these events registrations have been done manually. It is either through physical sign-up sheets or online forms with limited capabilities. These methods often lead to concerns such as fragmented participant data, a lack of confirmation for registrants, and difficulties in efficiently managing large volumes of registrations.

To solve these issues, a Student Event Registration System is proposed. This system is intended to simplify the registration process for student activities by offering a single platform for both event organizers and student participants. Organizers may create, amend and manage event information while students can quickly explore upcoming events, register and receive confirmation of attendance. The system also keeps track of users and events, making it simple to monitor past actions. This system uses object-oriented programming techniques to increase the efficiency, accuracy and convenience of handling student events in a university environment.

With an increasing number of student-led events each academic year, a reliable digital infrastructure is critical for scalability and user experience. The suggested solution not only streamlines registration and record-keeping but also improves communication between organizers and participants by providing real-time updates and notifications.

2.0 Problem statement

Universities often manage hundreds of student-led events annually, yet many still depend on disjointed manual or semi-digital registration methods such as paper forms, spreadsheets, or basic online surveys. These methods are not only time-consuming but also error-prone, leading to data inconsistencies, duplicated entries, and loss of participant information. Critically, these systems lack real-time validation, scalability, and integration with institutional records, making them unsuitable for high-volume or recurring event management.

This lack of centralized infrastructure creates operational friction. Students cannot access a unified event directory or verify their registration status instantly, while organizers must rely on manual quota tracking, participant filtering, and external tools for communication and updates. The absence of automation and feedback mechanisms not only reduces participation efficiency but also weakens the coordination between students and administrative units, ultimately affecting event quality and institutional engagement metrics.

Moreover, without a structured digital system, historical data on student involvement remains inaccessible, hindering meaningful event analytics and longitudinal tracking. As a result, institutions face limitations in evaluating event effectiveness, making data-driven decisions, or adapting future programs based on attendance trends. To ensure scalability, accuracy, and strategic insights, there is a critical need for a robust, object-oriented event registration system that centralizes management, supports real-time communication, and enhances overall user experience for both organizers and participants.

3.0 Class Diagram of thePproject

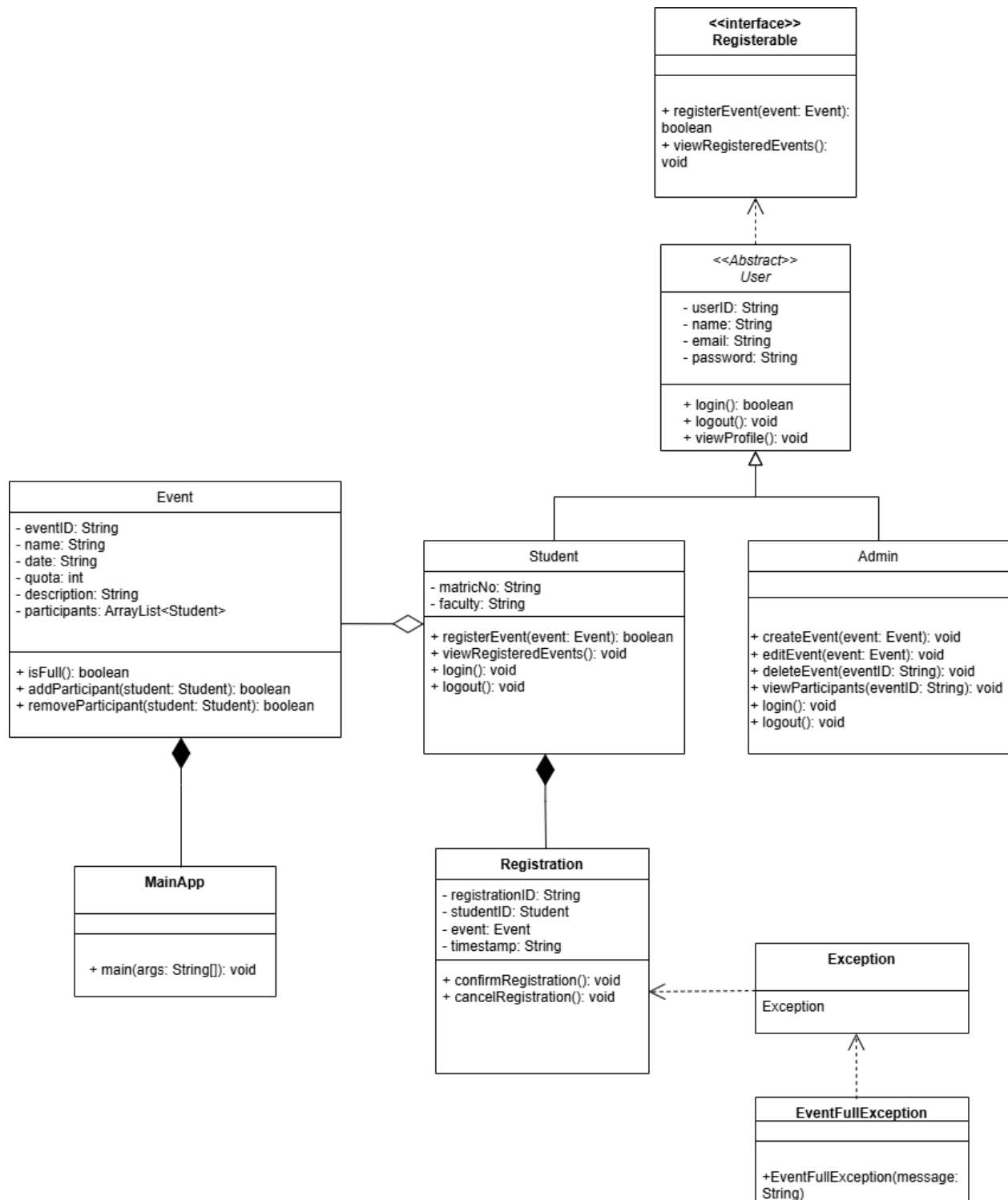


Figure 1: Class Diagram of Student Event Registration System

The table below shows the person in charge(PIC) in completing the coding of the classes.

Class	PIC
Registrable	Lau Yee Wen
User	Syikin
Student	Syikin
Admin	Syikin
Event	Ehtasham
Registration	Ehtasham
MainApp	Lau Yee Wen
EventFullException	Ehtasham

4.0 TECHNIQUES/SOFTWARE/TOOLS BEING USED

- ENUM
- ARRAY LIST
- ASSOCIATION
- INHERITANCE
- POLYMORPHISM
- ABSTRACT CLASS
- INTERFACE CLASS
- EXCEPTION HANDLING
- GUI APPLICATION